

Tide retrospective: the classic flat witness e^{-1/x^2}

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The flat-invisibility tide proved that *any* flat perturbation hides from all expansion orders, but the theorem quantifies over a hypothesis package: continuous, bounded, nonnegative, flat. The germbij note’s Proposition 4.1 discussion is anchored on a specific function — e^{-1/x^2} , the textbook example of flatness — and until a concrete instance satisfies the package in Lean, the formalised counterexample is conditional. This tide closes that gap. It was deferred from the parent tide by joint vote, with the proof route already supplied in the parent consult, so it is the cheapest kind of tide: pure instantiation, stacked on the parent branch, with the deliberation inherited.

2 The thing formalised

Eight declarations in `Laplace/OneD/FlatWitness.lean`, sorry-free with zero warnings. The definition extends e^{-1/x^2} by 0 at the origin; the load-bearing lemma is the *global* factorial bound¹

$$e^{-1/x^2} \leq n! x^{2n} \quad \text{for all } x \text{ and all } n,$$

from which nonnegativity, the bound by 1, continuity, and the flatness package all follow. The two instantiation corollaries² feed the witness to the parent theorems: perturbing the harmonic potential by e^{-1/x^2} shifts every compactly-supported-observable integral by $O(t^{-N})$ for every N , i.e. superpolynomially. Proposition 4.1’s counterexample is now witnessed, not just quantified.

3 Proof strategy

Everything reduces to one inequality. Setting $s = 1/x^2$, the single term of the exponential series gives $s^n/n! \leq e^s$ (`Real.sum_le_exp_of_nonneg` plus `Finset.single_le_sum`), and rearranging yields $e^{-s} \leq n!/s^n = n! x^{2n}$; in Lean the rearrangement goes through the product identity $x^{2n} s^n = 1$ rather than nested divisions. Continuity at the origin is a squeeze against the $n = 1$ case ($0 \leq f \leq x^2$); away from the origin it is composition. The numerical check (executed before formalisation) confirmed the bound on a grid, with the maximum ratio 0.3679 at $n = 1$ matching the analytic supremum $e^{-n} n^n / n!$ exactly.

¹`flatWitness.le_factorial_mul_pow`.

²`flat_witness_invisible` and `flat_witness_superpolynomial`.

4 Roadblocks and resolutions

Minor, all known classes. The catalogued cascading-rewrite gotcha struck again: rewriting $1 = e^0$ to compare $e^{-1/x^2} \leq 1$ also rewrote the 1 inside $1/x^2$, leaving `linarith` facing an opaque e^0/x^2 atom; the fix is a `calc` through e^0 that never rewrites the goal. `continuousAt_pow` takes its arguments as (x, n) , not (n, x) . One dead tactic after `field.simp`. Three fix cycles total, all catalogue lookups.

The genuine friction was infrastructure: `seed_worktree` failed on this host because the macOS system bash (3.2) has no associative arrays, so every worktree this session was silently falling back to `lake exe cache get`. The worktree was seeded manually by hardlinking from the parent tide's tree (`rsync --link-dest`), and the script was fixed portably in SRI (an `awk` lookup over manifest lines) and smoke-tested: ten packages, Mathlib included, now hardlink correctly.

5 The deliberation

No fresh consult. The parent tide's consult recommended exactly this follow-up and supplied the route:

For $f(x) = e^{-1/x^2}$ (extended by 0) one can in fact obtain the flat bound globally:
 $e^{-1/x^2} \leq n! x^{2n}$. [...] Thus one may take $C = n!$ and any positive δ .

Its advice to verify `Real.sum_le_exp_of_nonneg` rather than trust a guessed name was followed and correct: the lemma exists, and the guessed direct form (`pow_div_factorial_le_exp`) does not. This is the documented proceed-without-new-consult path: prior deliberation on this exact candidate plus an executed numerical check.

6 Mathlib vs new

Mathlib lookup: the exponential series bound, single-term extraction from a finite sum, the squeeze lemma, continuity combinators, and the eventually-nonzero neighborhood filter. New: only the assembly. Notably, Mathlib's own `expNegInvGlue` ($e^{-1/x}$ for $x > 0$) was not needed — the even variant is self-contained in 140 lines.

7 What the next tide inherits

The germbij Proposition 4.1 story is complete in Lean: general theorem, `Decay`-vocabulary packaging, and concrete witness. The remaining germbij commissions are the higher cumulant rungs of Theorem 3.1 (fifth/sixth cumulants, or the general-order induction) and the multivariate §7.4 recovery programme; both need fresh programme scoping before any tide starts.